DP Graded Quiz M7

Question 1

(**Note:** Use Python to solve this question). Use a Heston model with 20,000 simulations to price a European Put option with the following inputs:

- $\nu_0 = 0.095$
- Volatility rate of mean reversion $\kappa = 5$
- $\sigma = 0.35$
- Long-term average variance, $\theta = 0.1$
- Correlation $\rho = 0.25$
- $S_0 = 40.53$
- $K = 39$
- $r = 0.085$
- Number of time steps in the period: 100
- $T = 4/12$ (4 months)

Use a random seed equal to 42 (np.random.seed(42)). What is the price of the option?

- $2.04
- $4.65
- $1.67
- $3.12

Question 2

(**Note:** Use Python to solve this question.) Use a Merton model with 20,000 simulations to price a European put option with the following inputs:

- $\lambda = 0$
- $\mu = 0.5$
- $\delta = 0.75$
- $r = 0.05$
- $\sigma = 0.45$
- $T = 3$ months
- $S_0 = 200$
- $K = 195$
- Number of steps in period (for Merton): 90

With the same parameters (those that apply), do the same using Monte Carlo on a Black-Scholes framework with 1 time step. Use a random seed equal to 2 (np.random.seed(2))

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for both cases. What is the difference between the prices obtained?

- The Merton price is higher by $2.27.
- Absolute difference in prices is less than $0.50.
- The Merton price is lower by $1.15.
- (•) The Merton price is lower by $2.27.

Question 3

(**Note:** Use Python to solve this question.) Use the Heston model to do one simulation of the evolution of a stock price with the following inputs:

- $\nu_0 = 0.04$
- Volatility rate of mean reversion $\kappa = 2$
- $\sigma = 0.3$
- Long-term average variance, $\theta = 0.04$
- Correlation $\rho = -0.9$
- $S_0 = 100$
- $r = 0.05$
- Number of time steps in the period: 5
- Time, $T = 1$

Use a random seed equal to 42 (np.random.seed(42)). What is the final value for the volatility function you obtain for the last step in the period?

- 0.04
- 0.028
- 0.052
- (•) 99.06

Question 4

Given the following Merton model SDE:

$$dS_t = (r - r_j)S_t dt + \sigma S_t dZ_t + J_t S_t dN_t$$

where $r_j = \lambda \left(e^{\mu_j + \frac{\delta^2}{2}}\right) - 1$

- μ_j is the intensity of jumps
- μ_j is the average jump size
- λ is the average jump size

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- ◉ λ is a poisson random number

Question 5

(**Note:** Use Python to solve this question). Use a Heston model with 15,000 simulations to price a European Call option with the following inputs:

- $\nu_0 = 0.05$
- Volatility rate of mean reversion $\kappa = 1$
- $\sigma = 0.45$
- Long-term average variance, $\theta = 0.05$
- Correlation $\rho = -0.75$
- $S_0 = 35$
- $K = 33$
- $r = 0.05$
- Number of time steps in the period: 50
- $T = 3/12$ (3 months)

Use a random seed equal to 2 (np.random.seed(2)). What is the price of the option?

- ● $0.976
- ● $1.11
- ● $2.02
- ◉ $3.08

Question 6

(**Note:** Use Python to solve this question.) Use Merton model with 10,000 simulations to price a European put option with the following inputs:

- $\lambda = 0.75$
- $\mu = -0.5$
- $\delta = 0.25$
- $r = 0.05$
- $\sigma = 0.2$
- $T = 1$
- $S_0 = 50$
- $K = 40$
- Number of steps in period: 25

Use a random seed equal to 0 (np.random.seed(0)). What is the price of the option?

- ● $5.89

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- $2.63
- $3.88
- $16.29

Question 7

(**Note:** Use Python to solve this question.) Use the Heston model to do one simulation of the evolution of a stock price with the following inputs:

- $\nu_0 = 0.05$
- Volatility rate of mean reversion $\kappa = 1.25$
- $\sigma = 0.25$
- Long-term average variance, $\theta = 0.045$
- Correlation $\rho = -0.7$
- $S_0 = 50$
- $r = 0.02$
- Number of time steps in the period: 10
- Time, $T = 1$

Use a random seed equal to 5 (np.random.seed(5)). What is the final value for the stock price by the end of the period?

- $50
- $63.22
- $52.89
- $58.70

Question 8

Given the following Merton model SDE:

$$dS_t = (r - r_j)S_t dt + \sigma S_t dZ_t + J_t S_t dN_t$$

where

$$r_j = \lambda \left(e^{\mu_j + \frac{\delta^2}{2}} \right) - 1$$

A higher λ means that...

- There is a lower probability of experiencing a negative jump.
- There is a higher probability of experiencing a jump (positive or negative).
- There is a higher probability of experiencing a negative jump.

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- (●) There is a higher probability of experiencing a positive jump.

Question 9

The Heston model produces a distribution of the underlying asset returns...

- ( ) identical to CEV with constant volatility.
- ( ) closer to Black-Scholes than CEV.
- ( ) closer to reality than Black-Scholes.
- (●) almost identical to Black-Scholes.

Question 10

(**Note:** Use Python to solve this question.) Use a Merton model with 20,000 simulations to price a European call option with the following inputs:

- $\lambda = 0$
- $\mu = 0.5$
- $\delta = 0.75$
- $r = 0.05$
- $\sigma = 0.45$
- $T = 3$ months
- $S_0 = 200$
- $K = 195$
- Number of steps in period (for Merton): 90

With the same parameters (those that apply), do the same using Monte Carlo on a Black-Scholes framework with 1 time step. Use a random seed equal to 2 (np.random.seed(2)) for both cases. What is the difference between the prices obtained?

- ( ) The Merton price is higher by $1.54.
- ( ) The Merton price is lower by $2.27.
- ( ) Absolute difference in prices is less than $0.50.
- (●) The Merton price is lower by $1.54.

Question 11

(**Note:** Use Python to solve this question.) Use the Heston model to do 1 simulation of the evolution of a stock price with the following inputs:

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- $\nu_0 = 0.05$
- Volatility rate of mean reversion $\kappa = 1.25$
- $\sigma = 0.25$
- Long-term average variance, $\theta = 0.045$
- Correlation $\rho = -0.7$
- $S_0 = 50$
- $r = 0.02$
- Number of time steps in the period: 10
- Time, $T = 1$

Use a random seed equal to 0 (np.random.seed(0)). What is the final value for the volatility function you obtain for the last step in the period?

- 0.05
- 0.0069
- 0.0318
- $64.51

Time left: 2:59:33

QUESTIONS

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

Question 12

Compared to Black-Scholes, the Merton model can produce distributions...

- Closer to Normal
- With heavier tails
- Closer to Chi-square
- Closer to Lognormal

Question 13

(**Note:** Use Python to solve this question). Use a Heston model with 15,000 simulations to price a European Call option with the following inputs:

- $\nu_0 = 0.095$
- Volatility rate of mean reversion $\kappa = 5$
- $\sigma = 0.45$
- Long-term average variance, $\theta = 0.15$
- Correlation $\rho = -0.8$
- $S_0 = 325.3$
- $K = 330$
- $r = 0.05$
- Number of time steps in the period: 252
- $T = 1$ (1 year)

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Use a random seed equal to 2 (np.random.seed(2)). What is the price of the option?

- $66.23
- $51.95
- $43.15
- $4.67

Question 14

In a Merton model with $\mu_j < 0$...

- Mean returns will likely be higher.
- Return distribution will have a heavy left tail.
- Return distribution will exhibit no skewness.
- Return distribution will have a heavy right tail.

Question 15

Given the following SDEs for the variance of 4 Heston models...

Model 1:

$$d\nu(t) = 2.25\,(0.05 - \nu(t))dt + 0.25\sqrt{\nu(t)}dZ_2$$

Model 2:

$$d\nu(t) = 4.25\,(0.02 - \nu(t))dt + 0.75\sqrt{\nu(t)}dZ_2$$

Model 3:

$$d\nu(t) = 1.25\,(0.05 - \nu(t))dt + 0.75\sqrt{\nu(t)}dZ_2$$

Model 4:

$$d\nu(t) = 0.25\,(0.01 - \nu(t))dt + 0.95\sqrt{\nu(t)}dZ_2$$

In which model will volatility converge to its long-term mean more quickly?

- Model 4
- Model 1
- Model 2

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- (●) Model 3

Question 16

Based on the following discretized version of Merton model:

$$S_t = S_{t-1}\left(e^{\left(r-r_j-\frac{\sigma^2}{2}\right)dt+\sigma\sqrt{dt}z_t^1} + \left(e^{\mu_j+\delta z_t^2}-1\right)y_t\right)$$

What does z_t^2 represent?

- Standard normal random number for the drift process
- Standard normal random number for the diffusion process
- Poisson random number for the size of the jump
- (●) Standard normal random number for the size of the jump

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